STUDENT PERFORMANCE ANALYZER

Test Cases & Expected Results | BCA Project — PS No. 100

Project rules

Input file: **students.csv** | Subjects: Maths, Physics and Chemistry (each out of 100) | Percentage = Total / 3. Grades: A (≥90), B (≥75), C (≥60), D (≥40), F (<40).

Test Case 1 — Normal Input

Input CSV rows	Menu option	Expected output
Aman, 80, 70, 90 Riya, 60, 65, 70 Karan, 40, 35, 50	1 — Result	Aman | 240 | 80.00% | B Riya | 195 | 65.00% | C Karan | 125 | 41.67% | D

Test Case 2 — Boundary / Minimum Input

Input CSV rows	Menu option	Expected output
Neha, 100, 100, 100 Ravi, 0, 0, 0	1 — Result	Neha | 300 | 100.00% | A Ravi | 0 | 0.00% | F

Test Case 3 — Invalid Input

Input CSV row	Expected output
Amit, 85, abc, 70	Invalid marks for Amit in Physics: 'abc'. Row skipped.

Test Case 4 — Statistics (Same data as Test Case 1)

Input	Expected output
Aman 80,70,90 Riya 60,65,70 Karan 40,35,50 Menu option: 2	Mean %: 62.22 Median %: 65.00 Std Dev: 15.77 Topper: Aman (80.00%) Subject averages: Maths 60.00, Physics 56.67, Chemistry 70.00 Subject-average bar chart and percentage histogram displayed.

Note: The standard deviation expected by this assignment is the population standard deviation (NumPy `np.std()` default). Run each test by replacing the contents of students.csv with the listed input, then run the Python console program.